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GEOMAGNETIC POLARITY TRANSITIONS IN TWO ORIENTED SEDIMENT CORES FROM THE NORTHWEST PACIFIC

STEPHEN R. HAMMOND, STEFAN M. SEYB [1] and FRITZ THEYER

Hawaii Institute of Geophysics, University of Hawaii, Honolulu, HI 96822 (U.S.A.)

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The paths described by the geomagnetic poles during Jaramillo and Olduvai polarity transitions, as recorded in two oriented sediment cores from the northwest Pacific Ocean, closely resemble those previously determined in another deep-sea sediment core from the Indian Ocean. The paths determined for a specific transition appear to be the same in all three cores. There seems, however, to be a unique path associated with each of the three successive transitions studied in all three cores.

1. Introduction

Although it has been abundantly demonstrated that deep-sea sediments frequently preserve a reliable paleomagnetic record [1], until recently it has generally not been feasible to utilize paleomagnetic data from piston-cored deep-sea sediments for studies of polarity transitions. Such studies have been almost entirely dependent on subaerially exposed strata because, until the advent of very sensitive cryogenic magnetometers, the frequently encountered slow sediment accumulation rates and the usually relatively weak magnetization of deep-sea sediments both mitigated against the resolution of detailed field behavior. An additional important factor inhibiting such studies was the inability to record the in-situ orientation of sediments recovered by deep-sea coring or to ascertain what effects coring might have had on the recovered sediments. An instrument designed specifically to record corer behavior and orientation during sediment penetration has been successfully used to obtain this information and is now routinely used aboard Hawaii Institute of Geophysics research vessels [2]. The orienting device utilizes an 8-mm movie camera to record vertical and horizontal orientation, vertical acceleration, and time throughout free-fall and penetration.

The present work was undertaken to determine whether or not the paleomagnetism of oriented deep-sea sediment cores could provide additional insight into the nature of the geomagnetic field during periods of transition between normal and reversed polarity. Two fully oriented piston cores were recovered from an unnamed basin northeast of the Hess rise and southwest of the Emperor trough. Site locations for the two cores, K75-01 and K75-02, are 37°22.2′N, 179°36.0′W and 38°36.7′N, 179°20.9′E, respectively. Water depth at the K75-01 site is 5383 m and at the K75-02 site, 5748 m. Both cores are composed of occasionally mottled, but otherwise uniform, pelagic brown clays containing abundant sponge spicules, diatoms, and radiolarians as well as minor amounts of manganese micronodules.

The only other similar study of a deep-sea sediment core was by Opdyke et al. [3]. They recovered a core (RC14-14; 35°55′S, 59°58′E) with a high average rate of sediment accumulation from the Indian Ocean

Hawaii Institute of Geophysics Contribution.

[1] Present address: Mobil Exploration and Producing Services, Inc., P.O. Box 900, Dallas, TX 76221, U.S.A.

and were able to examine, in detail, the behavior of the geomagnetic field during the relatively brief intervals of time corresponding to the termination of the Olduvai and the onset and termination of the Jaramillo polarity events.

Relative orientation of the paleomagnetic samples from Opdyke et al.'s core was achieved by means of a continuous reference line that was scribed on the surface of the sediment during its extrusion from the core barrel. Subsequently, in order to determine virtual geomagnetic pole (VGP) paths during the polarity transitions, the means of the declination values in the intervals preceding each transition were adjusted to 180° or 360° depending on the polarity of the interval.

On the basis of the dissimilarity of the Olduvai and Jaramillo transition paths, Opdyke et al. [3] concluded that, while the field retained a small dipole component throughout the transitions, a non-dipole field was predominant during part of the transition period. They further concluded that decreases in remanent intensity, which were coincident with the polarity transitions in the core, were not caused by variations in magnetic mineralogy of the sediments but instead reflected actual decreases in the intensity of the earth's field. In the present study these same polarity transitions are investigated in two oriented sediment cores from a location in the northwest Pacific, approximately 120 longitudinal degrees away, and in the opposite hemisphere, from the Indian Ocean site.

2. Methods and magnetostratigraphy

The remanent magnetization of samples from cores K75-01 and K75-02 was measured with a Superconducting Technology, Inc. cryogenic magnetometer. The routine sampling interval was 10 cm but was decreased to 2 cm across the Jaramillo and Olduvai polarity transitions. Seven pilot samples from each core were subjected to stepwise partial demagnetization in alternating fields (AF) ranging from 25 to 500 Oe to selectively eliminate any viscous or secondary magnetization components. Symons and Stupavsky's [4] method, which was used to select the optimum of intensity for this purpose, yielded a minimum paleomagnetic stability index of 15 millidegrees per oersted at 100 Oe for K75-01 and at 150 Oe for K75-02. These respective partial demagnetization levels were subsequently applied to all samples. A close correspondence observed between the natural remanent magnetization (NRM) and the AF demagnetized values of remanent demagnetization, together with the results of the demagnetization analyses of the pilot samples, indicates that the sediments are magnetically stable. The stability of these sediments contrasts markedly with the highly unstable pelagic "red clays" found approximately 2° south of the present core sites [5,6]. The 95% confidence intervals for the final, partially demagnetized, inclination and declination data of each core (after all data were adjusted to normal polarity) are ±9.1° and ±19.1°, respectively, for core K75-01 and ±9.7° and ±24.7°, respectively, for K75-02.

With the exceptions of the uppermost 350 cm and lowermost 220 cm of K75-01, the magnetostratigraphic record in both cores is straightforward and easily interpreted (Fig. 1). (A relatively high water content, particularly within the upper portion of K75-01, resulted in this interval being deformed during handling and sampling procedures.) The sediments in K75-01 extend from the Brunhes to the upper Gauss epoch. Considering the average sediment accumulation rate prevailing during most of the Matuyama epoch (about 6 mm/10^3 years), and the fact that the core does not contain the Kaena event, the approximate age of the bottom of the core must be about 2.8 m.y. Although the interval between 1070 and 1150 cm in K75-01 was continuously sampled at 2-cm intervals, neither of the brief normal polarity events that are infrequently observed to precede and succeed the Olduvai event in deep-sea cores (the Reunion and Gilsa events, respectively) [1], was detected. Core K75-02 extends to the upper Matuyama epoch and an extrapolation of the sediment accumulation rate for the interval between the Brunhes-Matuyama boundary and the base of the Jaramillo event (17 mm/10^3 years) yields a basal age of about 1.1 m.y. Dating of both cores was independently achieved by means of their radiolarian and diatom biostratigraphies. The occurrence of *Rhizosolenia matuyamai* in each core, for example, specifically identifies the Jaramillo event [7].

The paleomagnetic declination data shown in Fig. 1 clearly preserve a record of polarity changes as evidenced by the 180° changes of the azimuthal

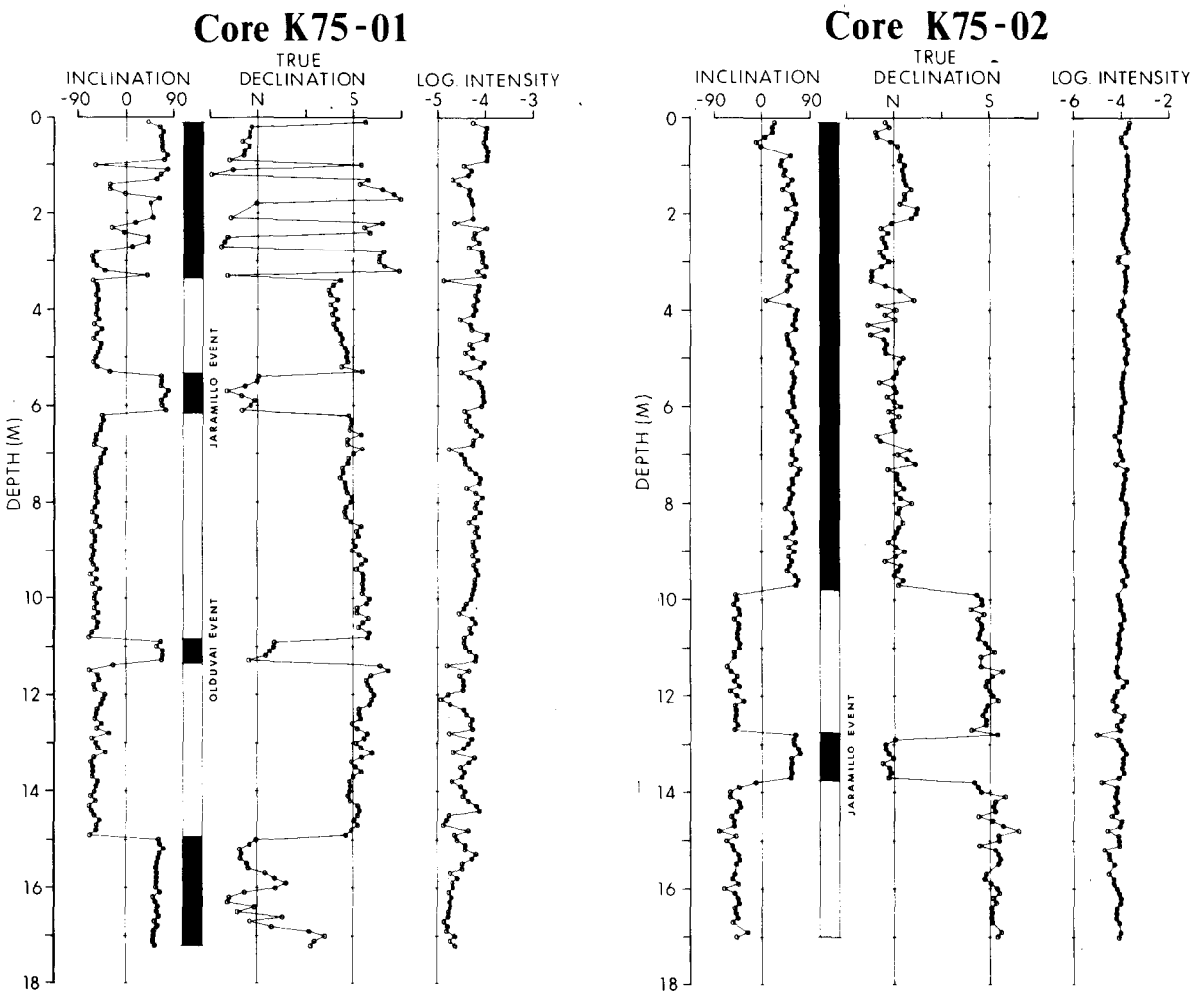


Fig. 1. Paleomagnetic data for cores K75-01 and K75-02 oriented with respect to true north and corrected for corer rotation. Solid intervals indicate normal polarity in the core; open intervals indicate reversed polarity.

vector direction, which are essentially synchronous with inclination sign changes. The declination data shown in the figure have been oriented with respect to true north (N) and were corrected for a progressive clockwise rotational offset in the unoriented data caused by rotation of the corer during sediment penetration. The origin of the slight but persistent deviation of the declination values from true north and south in K75-01, which remains after the effect of corer rotation has been removed, is unknown. The deviation may be the result of slippage, during coring, between the core liner and the sediment, caused by the aforementioned high water content of these sediments.

3. Results and discussion

3.1. Virtual geomagnetic poles

As an initial test of the oriented paleomagnetic data from cores K75-01 and K75-02, an average VGP position was determined for each core. To minimize scatter in VGPs, some data from the upper and lower portions of both cores (which are often disturbed during coring, e.g., the top and bottom of K75-01) were discarded. The data used to calculate the VGP locations included those within the intervals 350–1510 cm in K75-01 and 470–1680 cm in K75–02.

TABLE 1

Virtual geomagnetic poles and Fisher statistics for cores K75-01 and K75-02

Core	Pole latitude	Pole longitude	Number of points	k	α_{95} (°)	dp (°)	dm (°)
K75-01	86.9°N	107.3°W	116	45	2.2	2.4	3.2
K75-02	83.6°N	62.8°E	114	56	1.8	1.8	2.5

The resulting average VGPs, with the corresponding Fisher [8] statistics, are listed in Table 1 and the two poles with their 95% ovals of confidence are plotted in Fig. 2. If, as concluded from a paleomagnetic study of sediment cores from widespread locations throughout the world's oceans [9], the earth's magnetic field (when its configuration and orientation are averaged over long periods of time) closely approximates that of a geocentric axial dipole, then the locations of the average VGPs for K75-01 and K75-02 would not be expected to differ from the present geographic pole. The oval of confidence for K75-01 does encompass the geographic pole but the oval for the K75-02 pole is offset by about 4° at the 95% confidence level. The offsets of both VGPs are likely to be attributable to the degree of scatter not uncommonly observed in sediment paleomagnetic data. The relatively young age of the cores and the locations of their coring sites, relative to the Pacific plate poles of rotation for this time interval [10], preclude any discernible influence of plate motion on the location of either VGP. Non-geocentric geomagnetic field models, on the other hand, could account for locations of VGPs slightly offset from the geographic poles [11,12]. Although the present results are insufficient to allow a choice between alternative proposed field models, the acquisition of additional oriented deep-sea paleomagnetic data would make such a discrimination possible.

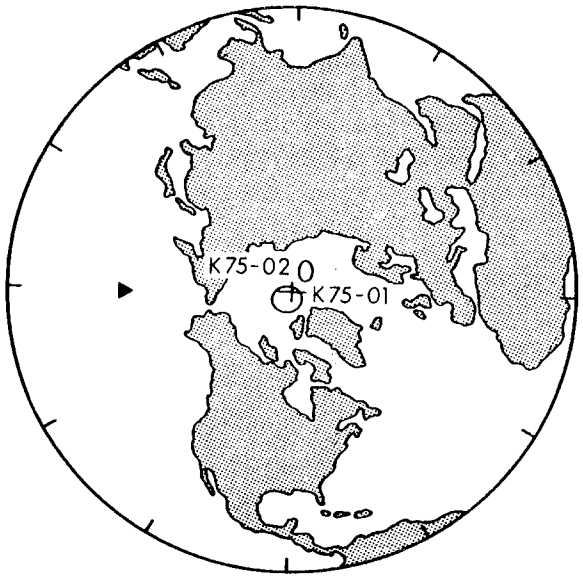
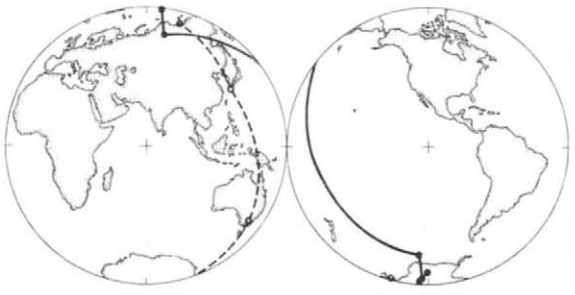


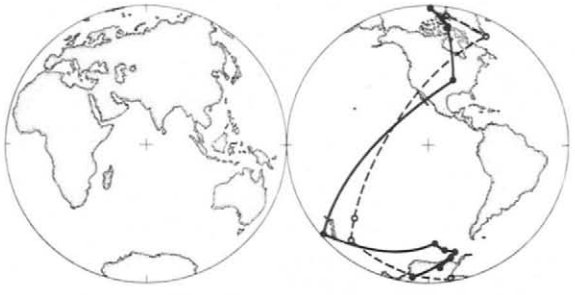
Fig. 2. Virtual geomagnetic poles for cores K75-01 and K75-02, and their associated 95% ovals of confidence (Table 1). The triangle encompasses the coring sites for both cores.

3.2. Polarity transitions

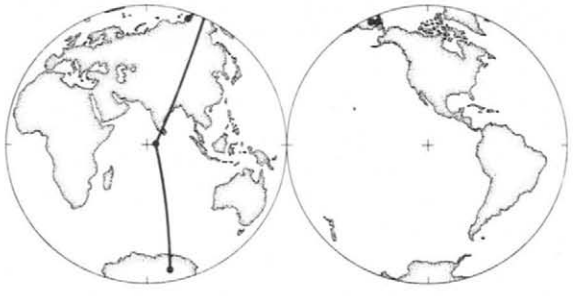
Between 0.92 and 0.87 m.y. B.P., the time of the Jaramillo event [1], the sediment accumulation rate for core K75-01 is about 16 mm/10^3 years and for core K75-02 about 20 mm 10^3 years. During the Olduvai event (1.86 to 1.71 m.y. B.P. [1]) the sediment accumulation rate in K75-01 is about 4 mm/10^3 years. Given the spacing of continuous paleomagnetic samples to be 2 cm, the high accumulation rate of core RC14-14 allowed Opdyke et al. [3] to sample at approximately 200-year intervals. For K75-01 and K75-02, the sampling interval during the Jaramillo event is about 1300 and 1000 years, respectively. Opdyke et al. [3] could, therefore, presumably detect field fluctuations, limited by the Nyquist frequency, with periods of 400 years, compared with periods of 2600 years for K75-01 and 2000 years for K75-02. Because of the apparently lower accumulation rate during the Olduvai interval in core K75-01, the shortest period of field fluctuation able to be resolved is approximately 10,000 years. For such applications, however, the Nyquist frequency yields only an



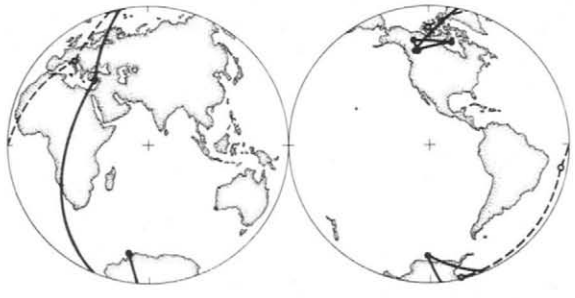
A K75-01 Onset of Jaramillo R→N



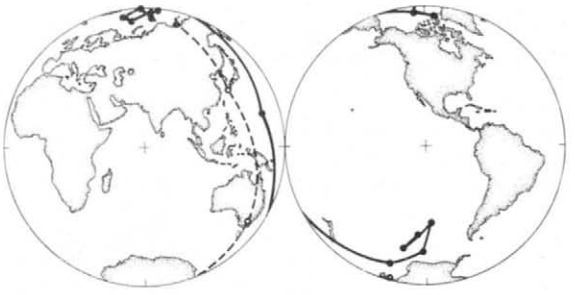
B K75-01 Termination of Jaramillo N→R



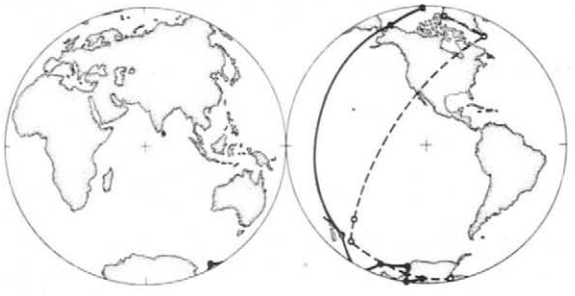
C K75-01 Onset of Olduvai R→N



D K75-01 Termination of Olduvai N→R



E K75-02 Onset of Jaramillo R→N



F K75-02 Termination of Jaramillo N→R

Fig. 3. Paths of virtual geomagnetic poles for four polarity transitions observed in K75-01 and two transitions observed in K75-02. Three of the four reversals were also studied by Opdyke et al. [3] in core RC14-14. Opdyke et al.'s [3] results are plotted as open circles and dashed lines for all reversals except the onset of the Olduvai (Fig. 3C).

approximation of the minimum detectable period, because the magnetic vector measured for each sample represents an average over the time span during which the sediment accumulated. Hence, the sampling is not strictly analogous to the discrete sampling used in conventional time series analysis.

With a single exception, at least one clearly intermediate transition direction was determined for each of the Jaramillo transitions in K75-01 and K75-02. Intermediate directions were also resolved for the Olduvai transitions in K75-01. The Jaramillo and Olduvai great circle transition paths for K75-01, and the Jaramillo transition paths for K75-02, are compared with those from RC14-14 in Fig. 3. The locations of the successive intermediate poles for K75-01 and K75-02 are given in Table 2. A comparison could not be made in Fig. 3C because the magnetostrati-

TABLE 2

VGP positions for transitions shown in Fig. 5

Transition	VGP
Core K75-01	
Termination of the Jaramillo event	81.7°S, 171.5°W
	68.7°S, 77.8°W
	77.3°S, 82.0°W
	64.9°S, 76.7°W
	65.0°S, 90.1°W
	60.7°S, 101.1°W
	41.2°S, 162.0°E
	38.1°N, 92.8°W
	74.4°N, 77.1°W
	88.2°N, 66.8°W
Onset of the Jaramillo event	83.0°N, 157.4°E
	69.0°N, 93.6°E
	66.9°S, 119.2°W
	84.9°S, 134.8°W
	85.1°S, 160.3°W
	79.4°S, 107.4°W
Termination of the Olduvai event	65.6°S, 45.1°E
	69.3°S, 112.1°W
	36.8°N, 32.4°E
	57.0°N, 122.9°W
	62.4°N, 85.8°N
	64.3°N, 127.0°W
	57.1°N, 120.3°W
Onset of the Olduvai event	71.8°N, 145.0°E
	67.7°N, 173.1°E
	66.1°N, 168.8°E
	00.4°N, 75.4°E
	76.1°S, 119.5°E
Core K75-02	
Termination of the Jaramillo event	85.2°S, 143.0°W
	81.3°S, 170.0°E
	74.2°S, 146.9°W
	67.1°S, 177.4°W
	60.9°S, 152.0°E
	44.8°S, 177.5°E
	71.2°N, 166.9°W
	89.1°N, 131.6°W
Onset of the Jaramillo event	83.8°N, 147.9°E
	84.6°N, 108.7°E
	78.8°N, 41.4°E
	79.2°N, 15.6°E
	88.5°N, 16.2°W
	81.7°N, 94.6°E
	84.2°N, 158.8°W
	82.4°N, 76.2°W
	15.2°N, 146.4°E
	68.3°S, 162.4°W
	64.8°S, 113.1°W
	46.0°S, 106.2°W
	62.1°S, 133.3°W
	53.1°S, 117.2°W

graphic record in K75-01 is the only one that includes the onset of the Olduvai event. Otherwise, there is a strong resemblance between VGP paths for the same transition in all three cores. The termination of the Olduvai reversal (Fig. 3D) in RC14-14 shows one intermediate point in the Mediterranean and one off the east coast of South America. The same reversal in K75-01 shows a VGP in the eastern Mediterranean. The relatively close correspondence between the VGP locations in the Mediterranean region suggests that this location on the K75-01 path actually represents an intermediate field direction even though, considering the average apparent accumulation rate for the Olduvai interval in K75-01 and the length of time required for a polarity transition (about 4600 years [3]), one would not expect to be able to resolve the Olduvai transition paths in this core. Possible explanations may be that the rate of sediment accumulation varied throughout the Olduvai interval in core K75-01 or that there is a brief undetected unconformity within the Olduvai interval. An observation that may support this latter possibility is that the apparent sediment accumulation rate during the Olduvai event is considerably slower than the average rate throughout the rest of the Matuyama epoch in this core.

The path corresponding to the termination of the Jaramillo event in RC14-14 (Fig. 3B or 3F) has two intermediate VGPs in the vicinity of New Zealand and a VGP path passing over central North America. Likewise, there are VGPs near New Zealand for this transition for both K75-01 and K75-02. The path corresponding to the onset of the Jaramillo in K75-02 (Fig. 3E) is in close agreement with that from RC14-14, as is, to a lesser extent, the same reversal from K75-01 (although the latter contains no clearly intermediate points). An equally significant observation that follows from the present results, as well as from those of Opdyke et al. [3], is that there is little, if any, correspondence between VGP paths for successive transitions.

4. Implications of results

In general terms, the hypotheses proposed to account for the origin of the earth's magnetic field fall into two categories distinguishable by their respective implications for behavior of the field

during polarity transitions. In one, the normally predominant dipole field decays substantially before reversing its polarity, thereby allowing the non-dipole field to predominate during a portion of the transition [13,14]. Implied characteristics of such a transition field are that (1) paleomagnetically determined paths of VGPs for the same polarity transition, as seen from differing localities, would be unlikely to resemble each other, because the magnetic effects of the non-dipole field are not the same worldwide, and (2) paths of VGPs for different polarity transitions, but observed from the same geographic location, would not necessarily repeat themselves either, because, barring the existence of persistent components of the non-dipole fields, the non-dipole configuration would be unlikely to repeat itself. Opdyke et al. [3] concluded that since successive VGP paths in RC14-14 were not the same and did not coincide with those previously reported by Creer and Ispir [15], the geomagnetic field during a portion of a transition interval is dominantly non-dipolar.

Creer and Ispir [15] previously proposed an alternative transition model that incorporated a partially diminished dipole field that inverted to the opposite polarity. They determined that several successive Tertiary VGP paths were coincident, despite the fact that the paths were derived from data gathered at widely scattered localities. Consequently, they concluded that the field remains essentially dipolar throughout polarity transitions. They further proposed that the geomagnetic field results from the interaction of three dipoles.

Resolution of these conflicting results is important because the nature of the transition field places constraints on any theories concerning the origin of the geomagnetic field. Despite the relatively few intermediate positions located at mid to low latitudes ($<45°$N or S) in the present results, the correspondence between paths for a specific polarity transition is unlikely to be coincidental. While these few results are insufficient to resolve the conflict between the foregoing observations and conclusions, the resemblance between the VGP paths for K75-01, K75-02, and RC14-14 for corresponding transitions, together with the fact that the cores are from widely separated regions, tentatively supports a transition model in which a dipolar field is predominant during each of the polarity reversals studied. On the other hand, successive VGP paths do not coincide, and neither Opdyke et al.'s [3] nor the present VGP paths match those determined by Creer and Ispir [15]. Thus, while transition fields seem to retain predominantly dipolar configurations, the dipole (or possibly, the non-dipole) field may incorporate large-scale magnetic features that can differ between successive reversals. The location of transition paths, for example, may be influenced by the interaction between a dominant axial dipole and other lesser, independent, dipoles [13,15].

In each of the transitions studied, there seems to be a decrease in remanence intensity associated with the change in direction of the magnetization vector (Fig. 4). Such a decrease is consistent with either near total decay of the dipole field or inversion of a weak dipole. It should be noted, however, that Løvlie [16] has shown that such variations in intensity of sediment magnetization can be the result of post-depositional alignment of magnetic grains. Moreover, the present paleomagnetic intensity data have not been normalized for effects of either varying quantity or varying magnetic mineralogy (either of which could account for the decreases).

If, however, it is assumed that a decrease in intensity of the geomagnetic field is responsible for the decrease in magnetization intensity of the K75-01 and K75-02 sediments that span the Olduvai and Jaramillo transitions and, further, that magnetization intensity is directly proportional to field intensity during (or shortly after) the time the sediments were deposited, then the magnitude of the intensity drops recorded during the K75-02 transitions very tentatively support the non-dipole transition model. The K75-02 intensity values decrease to about 10% of the before and after transition values, a level consistent with the present non-dipole field intensity throughout most of the Pacific basin region [17]. Because of a lack of resolution, the intensity change in the K75-01 data yield no useful information, in favor of or against, either the non-dipole or dipole transition model. Creer and Ispir [15] have suggested that various patterns of reversals of both the axial dipole or subsidiary dipoles could account for an intensity drop of perhaps two to five times the relative decrease shown for most of the K75-01 transitions.

There is disagreement as to whether or not, during polarity transitions, the decrease in geomagnetic field

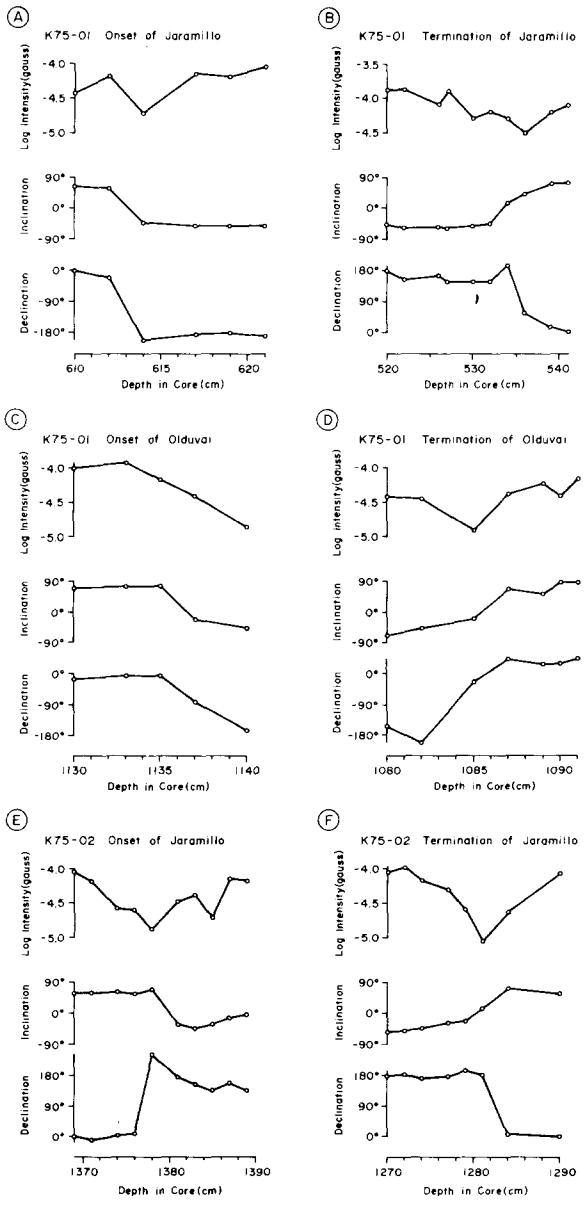


Fig. 4. Declination, inclination and intensity during the polarity transitions of K75-01 and K75-02 shown in Fig. 3.

intensity is coincident with the onset and termination of motion of the geomagnetic poles. Opdyke et al. [3] found that the intensity and VGP excursions were coincident in their RC14-14 data. Conversely, Dunn et al. [18] found that, in a Miocene reversal from a Mount Rainier lava sequence, the field intensity decreased by about one order of magnitude preceding the onset of magnetization direction changes. They estimated the duration of the intensity excursion to be about 10^4 years compared to a period of about 4000 years for a complete direction change. Similarly, intensity excursions have been reported to both precede and succeed direction changes in marine [19] and continental red bed sediments [20].

Despite the relatively low sediment accumulation rates in the present cores, the observed intensity decreases accompanying the polarity transitions in K75-01 and K75-02 exhibit a characteristic trend. As shown in Fig. 4, in all but one instance the intensity of magnetization begins to drop in advance of the directional change and recovers only after the directional change is complete. The exception is shown in Fig. 4, the onset of the Jaramillo in K75-01, where the intensity drop seems roughly coincident with the directional change. This core, however has a sediment accumulation rate about 20% lower during the Jaramillo than does core K75-02, hence this discrepancy may be due to inherently poorer resolution. Table 3 summarizes the approximate durations of the polarity transitions and the durations of the

TABLE 3

Approximate maximum time durations of directional changes and intensity excursions for reversals in cores K75-01 and K75-02

Reversal events	Duration of directional change (years)	Duration of intensity excursion (years)
K75-01 Onset of Jaramillo	2500	3000
K75-01 Termination of Jaramillo	2500	~20,000
K75-01 Onset of Olduvai	3000	~18,000
K75-01 Termination of Olduvai	3000	4000
K75-02 Onset of Jaramillo	3000	9000
K75-02 Termination of Jaramillo	3000	9000

associated intensity excursions for the reversals in cores K75-01 and K75-02 based on the calculated sediment accumulation rates. The present results (subject to the assumptions discussed earlier) appear to indicate that the intensity change takes place over a longer period of time than does the directional change. The eventual resolution of this conflict may provide additional parameters that will help distinguish between dynamo models [14].

5. Conclusion

With paleomagnetic results from two fully oriented deep-sea cores from the northwest Pacific Ocean, two core-averaged virtual geomagnetic poles were determined that are close to the present geographic pole. Agreement between corresponding VGP paths associated with Jaramillo and Olduvai polarity transitions in these two cores and another core from the Indian Ocean is consistent with the existence of a dipolar geomagnetic field during these transitions. The apparent dissimilarity between successive transition paths, however, suggests that conditions within the earth's core that affect the path of the field during polarity transitions may vary over periods of time as short as the duration of the Jaramillo event (50,000 years).

Acknowledgements

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References

1 N.D. Opdyke, Paleomagnetism of deep-sea cores, Rev. Geophys. Space Phys. 10 (1972) 213.
2 S.M. Seyb, S.R. Hammond and T. Gilliard, A new device for recording the behavior of a piston corer, Deep-Sea Res. 24 (1977) 943.
3 N. D. Opdyke, D.V. Kent and W. Lowrie, Details of magnetic polarity transitions recorded in a high deposition rate deep sea core, Earth Planet. Sci. Lett. 20 (1973) 315.
4 D.T.A. Symons and M. Stupavsky, A rational paleomagnetic stability index, J. Geophys. Res. 79 (1974) 1718.
5 N.D. Opdyke and J.H. Foster, Paleomagnetics of cores from the North Pacific, in: Geological Investigations of the North Pacific, J.D. Hays, ed., Geol. Soc. Am. Mem. 126 (1970) 83.
6 D.V. Kent and W. Lowrie, Origin of magnetic instability in sediment cores from the central North Pacific, J. Geophys. Res. 79 (1974) 2987.
7 L.H. Burckle, S.R. Hammond and S. Seyb, A stratigraphically important new diatom from the Pleistocene of the North Pacific, Pac. Sci. 32 (1978) 209.
8 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London 217 (1953) 295.
9 N.D. Opdyke and K.W. Henry, A test of the dipole hypothesis, Earth Planet. Sci. Lett. 6 (1969) 139.
10 S.R. Hammond, D. Epp and F. Theyer, Neogene relative motion between the Pacific plate, the mantle, and the earth's spin axis, Nature (in press).
11 R.L. Wilson, Permanent aspects of the earth's non-dipole magnetic field over upper Tertiary times, Geophys. J. 19 (1970) 417.
12 N.D. Watkins, Hemispherical contrast in support for the offset dipole hypothesis during the Brunhes epoch: the case for an unequal co-axial dipole pair as a possible geomagnetic field source, Geophys. J.R. Astron. Soc. 28 (1972) 193.
13 P. Dagley and E. Lawley, Paleomagnetic evidence for th transitional behaviour of the geomagnetic field, Geophy J.R. Astron. Soc. 36 (1974) 577.
14 A. Cox, J. Hillhouse and M. Fuller, Paleomagnetic recor of polarity transitions, excursions, and secular variation, Rev. Geophys. Space Phys. 13 (1975) 185.
15 K.M. Creer and Y. Ispir, An interpretation of the behaviour of the geomagnetic field during polarity trans tions, Phys. Earth Planet. Inter. 2 (1970) 293.
16 R. Løvlie, The intensity pattern of post-depositional remanence acquired in some marine sediments deposited during a reversal of the external magnetic field, Earth Planet. Sci. Lett. 30 (1976) 209.
17 R.R. Doell and A. Cox, the Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in: The Nature of the Solid Earth, E.C. Robertson, ed. (McGraw-Hill, New York, N.Y.), 245.
18 J.R. Dunn, M. Fuller, H. Ito and V.A. Schmidt, Paleomagnetic study of a reversal of the earth's magnetic field, Science 172 (1971) 840.
19 C.R. Denham, R.F. Anderson and M.P. Bacon, Paleomagnetism and radiochemical age estimates for late Brunhes polarity episodes, Earth Planet. Sci. Lett. 35 (1977) 384.
20 C.G. Baag and C.E. Helsley, Penecontemporaneous magnetization of Moenkopi formation, J. Geophys. Res. 79 (1974) 3308.